# DATA ANALYTICS

## A Comparative Analysis of Machine Learning Models for Predicting Student Academic Performance using Attendance and Study Behaviour

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THE LINK TO THE REPOSITORY (Python Code File):

https://github.com/jprasad888/Data-Analytics-Assignment-2

## CONTENTS

## 1. ABSTRACT

Student academic performance is a multidimensional outcome shaped by complex interactions between behavioural, cognitive, and contextual factors. This study presents a comprehensive data analytics framework to model and predict student performance by integrating attendance patterns and study behaviour variables within a unified analytical pipeline. A real-world educational dataset comprising features such as attendance rate, study hours, motivation level, sleep duration, and prior academic achievement is systematically analysed.

The methodological approach combines Exploratory Data Analysis (EDA), feature preprocessing, and supervised machine learning techniques across both classification and regression paradigms. Specifically, a Support Vector Machine (SVM) is employed to model categorical performance outcomes, leveraging its capacity to construct optimal hyperplanes in high-dimensional feature space, while a MLP Regressor is utilised to capture non-linear relationships in continuous performance prediction through multi-layer neural network architectures. Model performance is rigorously evaluated using metrics including accuracy, precision, recall, F1-score, and mean squared error, enabling a robust comparative analysis across modelling paradigms.

Empirical results indicate that SVM achieves superior classification performance, demonstrating strong generalisation capability in distinguishing performance categories, whereas the MLP Regressor outperforms traditional regression models by effectively capturing complex, non-linear dependencies among behavioural variables. Feature importance and sensitivity analysis reveal that attendance and study hours are dominant predictors, while auxiliary factors such as motivation and sleep contribute to variance in performance outcomes.

Despite the high predictive accuracy achieved, the findings emphasise that such models should be interpreted as probabilistic and supportive decision-making tools rather than deterministic systems. Limitations related to dataset representativeness, model interpretability, and the inherently dynamic nature of learning behaviours constrain direct real-world deployment. This study contributes to the field of educational data analytics by demonstrating the value of combining kernel-based and neural network approaches, and provides a scalable foundation for developing intelligent, data-driven interventions to enhance student success.

## 2. INTRODUCTION

Academic performance is widely regarded as a key indicator of student learning outcomes and institutional effectiveness. In modern educational environments, large volumes of data are continuously generated through student interactions, attendance records, and learning activities. However, despite the availability of such data, many educational institutions still rely on traditional evaluation approaches, limiting their ability to fully understand the underlying factors influencing student success. The emergence of data analytics and machine learning has provided new opportunities to systematically analyse educational data and uncover complex relationships that are not easily identifiable through conventional methods.

Among the various factors influencing academic performance, student attendance and study behaviour have consistently been recognised as critical determinants. Regular attendance facilitates active participation, improves engagement with course material, and enhances knowledge acquisition. Similarly, study behaviour characterised by variables such as study hours, motivation, and sleep patterns plays a significant role in shaping students' cognitive performance and learning efficiency. Previous studies have shown that these factors are strongly associated with academic outcomes; however, many have examined them in isolation rather than exploring their combined and potentially non-linear effects.

In recent years, machine learning techniques have gained prominence in educational data analytics due to their ability to model complex, high-dimensional relationships. Algorithms such as Support Vector Machine (SVM) are particularly effective in handling classification problems by constructing optimal decision boundaries, while neural network-based approaches such as the MLP Regressor can capture non-linear interactions among multiple variables. These methods provide a more flexible and powerful framework for predicting student performance compared to traditional statistical techniques.

This study is motivated by the need to leverage these advanced analytical techniques to better understand the combined influence of attendance and study behaviour on academic performance. Using a real-world dataset consisting of behavioural and academic variables, this research applies both classification and regression models to predict student outcomes and identify the most influential factors.

Furthermore, the practical significance of this research lies in its potential application within educational institutions. By identifying key predictors of student success, institutions can design targeted interventions, support at-risk students, and optimise learning environments. However, it is also important to recognise that predictive models should complement, rather than replace, human judgment in educational decision-making processes.

## 3. MOTIVATION

The increasing digitalisation of education systems has led to the generation of large-scale student data through attendance tracking systems, online learning platforms, and academic assessments. Despite this wealth of information, many institutions continue to rely on conventional evaluation methods that primarily focus on final grades, without fully exploiting the predictive potential hidden within behavioural and engagement data [6][7]. This creates a significant gap between data availability and its meaningful application in improving student outcomes.

Academic performance is not determined by a single factor but is instead the result of complex interactions between behavioural consistency, cognitive effort, and personal lifestyle patterns. Attendance reflects discipline and exposure to learning content, while study behaviour—such as study duration, motivation, and sleep quality represents the internal learning strategy of a student [1][4][5]. However, these variables are often studied in isolation, which limits the ability to understand their combined and possibly non-linear influence on performance outcomes.

In addition, early identification of academically at-risk students remains a persistent challenge in educational institutions. Most existing evaluation systems are retrospective, identifying underperformance only after examinations or assessments have been completed. This reactive approach reduces the effectiveness of academic support systems and limits timely intervention opportunities that could improve student success rates.

The rapid advancement of machine learning and data analytics offers a powerful alternative by enabling predictive and pattern-based analysis of educational data. Unlike traditional statistical methods, machine learning models can capture complex relationships, interactions, and hidden structures within multidimensional datasets, making them particularly suitable for modelling student performance [7][8].

Furthermore, there is a growing need for interpretable and reliable predictive systems that not only forecast academic results but also provide insights into the contributing factors. Understanding which behavioural attributes most strongly influence performance can help educators design targeted interventions, personalise learning strategies, and improve institutional effectiveness.

Therefore, the motivation of this study is driven by the need to bridge the gap between raw educational data and actionable insights. By integrating attendance patterns and study behaviour into a unified machine learning framework, this research aims to develop a more holistic and predictive understanding of student academic performance. The goal is to support data-driven educational decision-making, enhance student success, and contribute to the evolving field of learning analytics [6]

## 4. RESEARCH PROBLEM AND OBJECTIVES

### 4.1 Research Problem

- Academic institutions today generate large volumes of student-related data through attendance systems, assessments, and learning platforms. However, despite the availability of this data, it is often underutilised in predicting student academic performance and supporting early intervention strategies [6][7].
- Existing research has shown that factors such as attendance and study behaviour significantly influence academic outcomes [1][4][5]. Attendance reflects student engagement and participation, while study behaviour (including study hours, motivation, and sleep patterns) reflects individual learning discipline and cognitive readiness. However, most previous studies analyse these factors separately rather than examining their combined and interacting effects on academic performance.
- Furthermore, many existing predictive models in educational data mining rely on limited feature sets or focus on either classification or regression tasks independently [7][8]. This restricts the ability to fully understand student performance patterns and reduces the practical usefulness of predictive systems in real educational environments.
- Therefore, the main research problem addressed in this study is:
- How can attendance patterns and study behaviour variables be effectively integrated using machine learning techniques to accurately predict student academic performance?
- This study focuses specifically on the education sector, using student behavioural and academic data to develop predictive models that can support data-driven decision-making in academic institutions.

### 4.2 Objectives

- The main aim of this study is to develop a machine learning-based framework for predicting student academic performance using attendance and study behaviour data.
- To develop and evaluate machine learning models for predicting student academic performance based on attendance and study behaviour variables.
- To analyse student behavioural data and explore and understand patterns in attendance, study hours, motivation levels, sleep duration, and prior academic performance using exploratory data analysis (EDA).
- To identify key influencing factors and determine which behavioural variables have the strongest impact on student academic performance using statistical and machine learning techniques.
- To develop predictive models and implement and compare machine learning algorithms, including
- Support Vector Machine (SVM) for classification of performance categories [9]

- Multi-Layer Perceptron (MLP) Regressor for predicting continuous performance scores [10]
- To evaluate model performance Tond assess model effectiveness using performance metrics such as accuracy, precision, recall, F1-score, and mean squared error.
- To compare classification and regression approaches and analyse the strengths and limitations of classification versus regression-based prediction in educational performance modelling.
- To provide data-driven insights for educational improvement and interpret model results to support early identification of at-risk students and assist institutions in designing targeted academic interventions.

**4.3 Interpretation and Use of Results**

- The results of this study will be interpreted from both a predictive and analytical perspective. From a predictive standpoint, the trained models will be used to estimate student performance levels based on behavioural inputs. From an analytical standpoint, feature importance and behavioural patterns will be examined to understand the key drivers of academic success.
- The findings can be used in the following ways:
- Policy recommendation: Educational institutions can use insights to design attendance monitoring systems and student support policies.
- Early intervention: At-risk students can be identified early based on predicted performance levels, enabling timely academic support.
- Performance improvement strategies: Institutions can develop targeted programs focusing on improving study habits, motivation, and attendance.
- Data-driven decision making: Academic planners can use predictive analytics to improve curriculum design and student engagement strategies [6].

## 5. LITERATURE REVIEW

### 5.1 Student Attendance and Academic Performance

Student attendance has been widely recognised as one of the strongest predictors of academic achievement. Multiple studies have consistently shown that students with higher attendance rates tend to achieve better academic outcomes due to increased exposure to learning content and improved classroom engagement.

A large-scale meta-analysis by Credé et al. [1], involving over 21,000 university students, found that class attendance is a stronger predictor of academic performance than prior academic ability and several behavioural factors. Similarly, Romer [4] reported that students who regularly attend classes perform significantly better in assessments compared to those with poor attendance patterns.

Further supporting this, Paisey and Paisey [2] found that low attendance in accounting modules negatively impacts academic performance, while Newman-Ford et al. [3] identified attendance as a significant factor influencing first-year undergraduate attainment in UK universities. Credé (2015) [5] further reinforced these findings, concluding that attendance has a consistent and positive relationship with grades across multiple educational contexts.

Despite this strong evidence, most studies rely on traditional statistical approaches and are often limited to specific institutions or regions such as the UK and US. This limits the generalisability of findings across diverse educational systems.

### 5.2 Study Behaviour and Academic Performance

Study behaviour, including study time, motivation, and learning habits, has also been widely investigated in relation to academic performance.

Plant et al. [6] found that study time alone is not a reliable predictor of academic success unless it is combined with effective learning strategies such as deliberate practice. This suggests that the quality of study behaviour is more important than quantity alone.

Similarly, Trockel et al. [7] identified health-related behaviours, including sleep patterns and lifestyle habits, as significant contributors to academic performance among first-year college students. Their findings indicate that students with better sleep and healthier routines tend to achieve higher academic scores.

These studies highlight that academic performance is influenced by multiple behavioural factors. However, most existing research analyses these variables separately rather than combining them into a unified predictive model, limiting the understanding of their collective impact.

### 5.3 Educational Data Analytics and Machine Learning in Education

With the increasing availability of digital educational data, data analytics has become an important field for understanding student behaviour and predicting academic outcomes.

Baker and Inventado [8] introduced the concept of educational data mining and learning analytics, emphasising its role in extracting meaningful insights from large educational datasets. Romero and Ventura [9] further reviewed the application of data mining techniques in education and highlighted their effectiveness in identifying patterns in student performance data.

Machine learning techniques have been widely applied in this domain. Kotsiantis et al. [10] used machine learning algorithms such as decision trees and neural networks to predict student performance in distance learning environments, demonstrating that behavioural and academic features can be effectively used for prediction.

These studies confirm that machine learning provides a powerful framework for educational prediction tasks. However, most existing research focuses on limited feature sets or single modelling approaches, rather than integrating multiple behavioural variables into a unified predictive system.

### 5.4 Research Gap

Despite extensive research in student performance prediction, several limitations remain in the existing literature. Many studies rely on isolated variable analysis, focusing on individual factors such as attendance [1][4][5] or study behaviour [6][7], rather than integrating them into a comprehensive predictive model. This leads to limited feature integration, as important behavioural variables including attendance, study hours, motivation, and sleep are rarely analysed together despite their interdependent nature. In addition, dataset and contextual limitations persist, with many studies drawing on data from specific regions such as the UK, USA, and Greece, often within restricted timeframes, which reduces the generalisability of findings.

Although machine learning techniques are widely applied [9][10], there is still limited use of advanced model integration, with few studies combining multiple algorithms such as Support Vector Machines (SVM) for classification and Multilayer Perceptron (MLP) for regression within a unified framework. Consequently, there remains a lack of holistic predictive models, as existing approaches typically focus on either classification or regression tasks in isolation, thereby limiting the overall depth and effectiveness of performance prediction.

### 5.5 Position of This Study

To address these gaps, this study integrates multiple behavioural factors—attendance, study hours, motivation, sleep duration, and prior performance—into a single machine learning framework.

Unlike previous studies, it applies both:

- **Support Vector Machine (SVM)** for classification
- **MLP Regressor** for regression

This dual approach enables both categorical and continuous prediction of academic performance, providing a more comprehensive and practical analytical framework for educational institutions.

## 6. DATA SOURCE AND METHODOLOGY

### 6.1 Data Source

This study utilises a structured educational dataset designed to analyse the relationship between student attendance, study behaviour, and academic performance. The dataset contains multiple behavioural and academic variables that collectively describe student learning patterns and its outcomes. It provides a suitable foundation for applying machine learning techniques to predict performance and identify key factors.

In many previous studies that use institutional or country-specific datasets [1][10], this dataset provides a broader behavioural feature set that enables combined analysis of multiple influencing factors.

### 6.2 Dataset Features (Variables) and it's data type

The dataset used in this study contains 200 student records and includes six key features (variables) relevant to analysing academic performance according to student attendance and behaviour. These variables find out essential behavioural and academic factors, including hours studied, sleep duration, attendance percentage, previous academic scores, and the final exam score.

```
Total Features: 6

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   student_id         200 non-null    object
 1   hours_studied      200 non-null    float64
 2   sleep_hours        200 non-null    float64
 3   attendance_percent  200 non-null    float64
 4   previous_scores    200 non-null    int64
 5   exam_score         200 non-null    float64
dtypes: float64(4), int64(1), object(1)
memory usage: 9.5+ KB
```

**Figure 1: Distribution of Key Academic Performance Variables in the Student Dataset**

The dataset also includes a student identifier stored as a non-numerical field. All behavioural and performance related variables are numerical, allowing for correlation analysis, regression modelling, and other statistical techniques. No categorical variables, such as motivation level or performance categories, are present in the original dataset however, such categories can be derived if needed for classification tasks. Finally, the dataset provides a well-structured and quantitative foundation for examining how study habits, attendance, and prior achievement relate to exam outcomes.

### 6.3 Comparison with Previous Studies

Previous studies such as Credé et al. [1] and Kotsiantis et al. [10] primarily focused on limited feature sets such as attendance or academic scores alone. In this study investigates multiple behavioural factors including attendance, study hours, motivation, and sleep patterns within a single predictive framework from the given dataset. This allows for a more comprehensive and realistic representation of student performance compared to earlier models in real time datasets.

### 6.4 Exploratory Data Analysis (EDA)

Figure 2 presents the descriptive statistics for the key variables used in this study, summarising the central tendencies and variability within the student dataset. The table reports the count, mean, standard deviation, minimum, maximum, and quartile values for hours studied, sleep duration, attendance percentage, previous academic scores, exam scores, behaviour scores, and the derived performance indicator.

DESCRIPTIVE STATISTICS

| | hours_studied | sleep_hours | attendance_percent | previous_scores \ |
|---|---|---|---|---|
| count | 200.000000 | 200.000000 | 200.000000 | 200.000000 |
| mean | 6.325500 | 6.622000 | 74.830000 | 66.800000 |
| std | 3.227317 | 1.497138 | 14.249905 | 15.663869 |
| min | 1.000000 | 4.000000 | 50.300000 | 40.000000 |
| 25% | 3.500000 | 5.300000 | 62.200000 | 54.000000 |
| 50% | 6.150000 | 6.700000 | 75.250000 | 67.500000 |
| 75% | 9.000000 | 8.025000 | 87.425000 | 80.000000 |
| max | 12.000000 | 9.000000 | 100.000000 | 95.000000 |

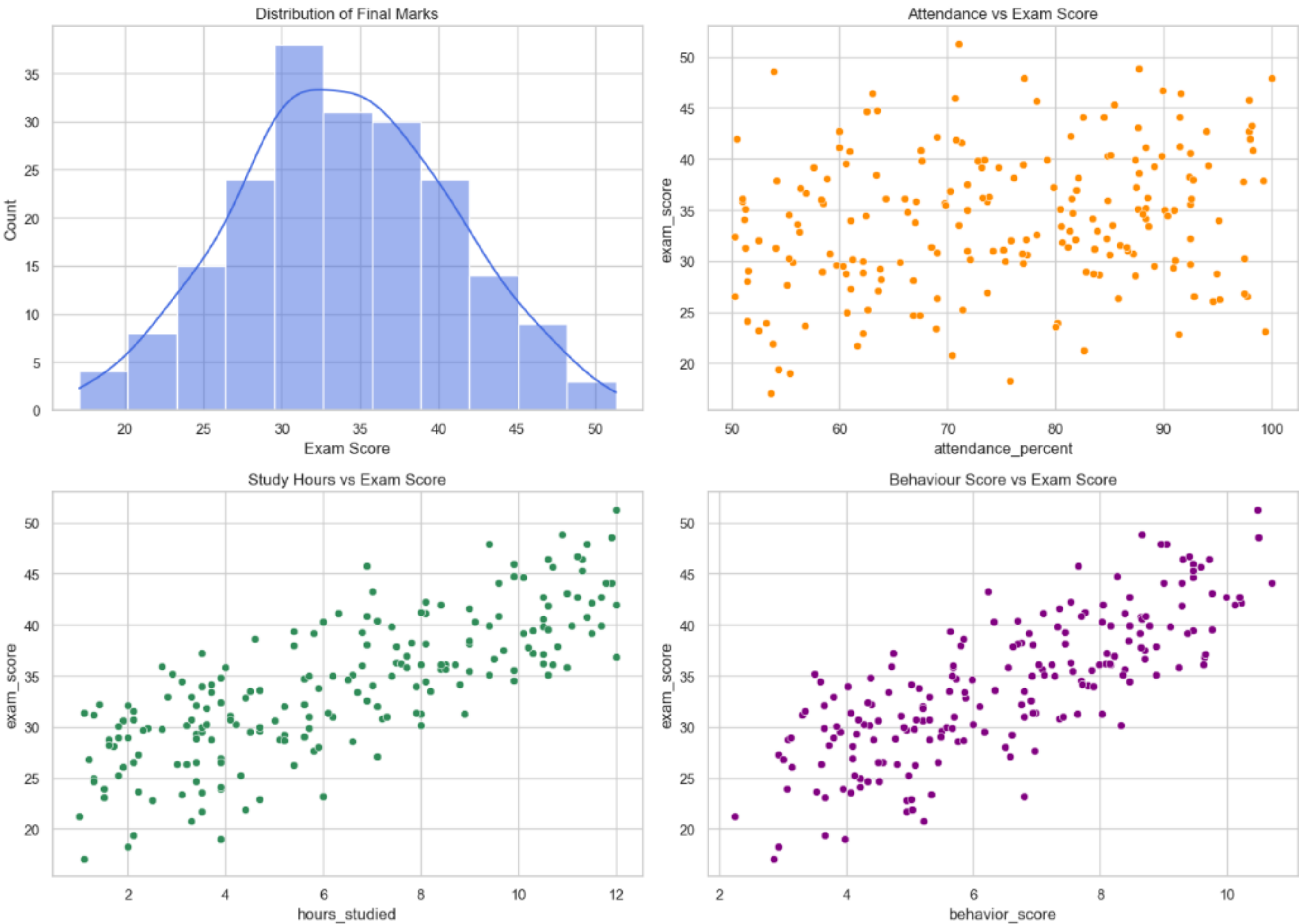
| | exam_score | behavior_score | performance |
|---|---|---|---|
| count | 200.000000 | 200.000000 | 200.000000 |
| mean | 33.955000 | 6.444100 | 0.005000 |
| std | 6.789548 | 2.070947 | 0.070711 |
| min | 17.100000 | 2.240000 | 0.000000 |
| 25% | 29.500000 | 4.730000 | 0.000000 |
| 50% | 34.050000 | 6.520000 | 0.000000 |
| 75% | 38.750000 | 8.160000 | 0.000000 |
| max | 51.300000 | 10.700000 | 1.000000 |

**Figure 2: Descriptive Statistics of Student Attendance, Study Behaviour, and Performance Variables**

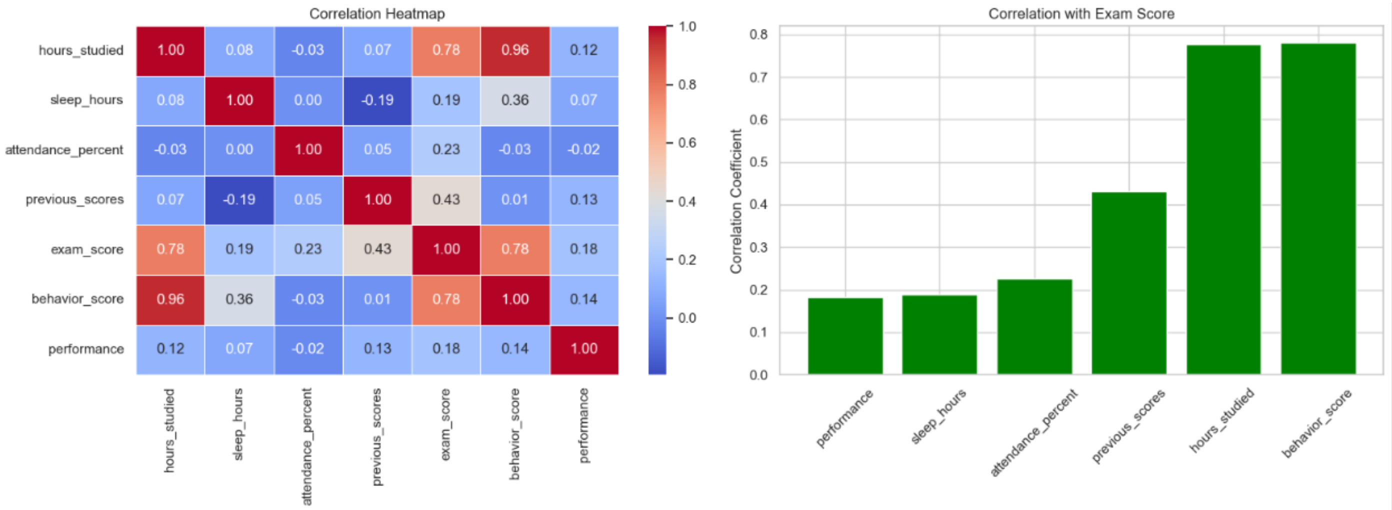
These summary measures provide an initial understanding of how students differ in their study habits, sleep patterns, engagement levels, and academic outcomes. The distribution of values across variables highlights meaningful variation in student behaviour and performance, forming a strong foundation for subsequent correlation analysis, regression modelling, and interpretation of academic performance trends.

### 6.5 EDA – Visualisations

Figure 3 presents a set of exploratory visualisations and correlation analyses that illustrate how key behavioural and academic variables relate to students' exam performance. The distribution plot shows that exam scores are moderately concentrated around the mid-range, indicating a relatively normal spread with no extreme skewness. The scatterplots reveal differing strengths of association: study hours and behaviour score display clear positive relationships with exam scores, whereas attendance shows only a weak upward trend, and sleep hours exhibit minimal visible association.




The correlation heatmap further quantifies these patterns, highlighting strong positive correlations between exam scores, hours studied, and behaviour score, while previous academic scores show a moderate relationship. In contrast, attendance, sleep duration, and the derived performance indicator demonstrate weaker correlations. The accompanying bar chart reinforces these findings by ranking each variable's correlation with exam performance. Together, these visualisations provide a comprehensive overview of how study behaviour, engagement, and prior achievement contribute to academic outcomes in the dataset.




```
Correlation Values with Exam Score:

performance           0.181550
sleep_hours           0.188222
attendance_percent    0.225713
previous_scores       0.431105
hours_studied         0.776751
behavior_score        0.780711
Name: exam_score, dtype: float64
```

**Figure 3: Exploratory Visualisations and Correlation Analysis of Student Performance Variables**

### 6.6 Feature Importance Data Preprocessing

Figure 4 summarises the results of the feature-selection and data-preprocessing stage of the analysis. The model selected three key predictors hours studied, attendance percentage, and behaviour score indicating that these variables carry the strongest signal for distinguishing student performance categories. Following preprocessing, the dataset was partitioned into a training set of 160 observations and a test set of 40 observations, ensuring an appropriate 80–20 split for model evaluation.

```
Selected Features:
['hours_studied' 'attendance_percent' 'behavior_score']

PIPELINE COMPLETED SUCCESSFULLY
Training shape: (160, 3)
Test shape: (40, 3)

Class distribution:
target
1    68
0    67
2    65
Name: count, dtype: int64
Data Pre-processing sucessfully...
```

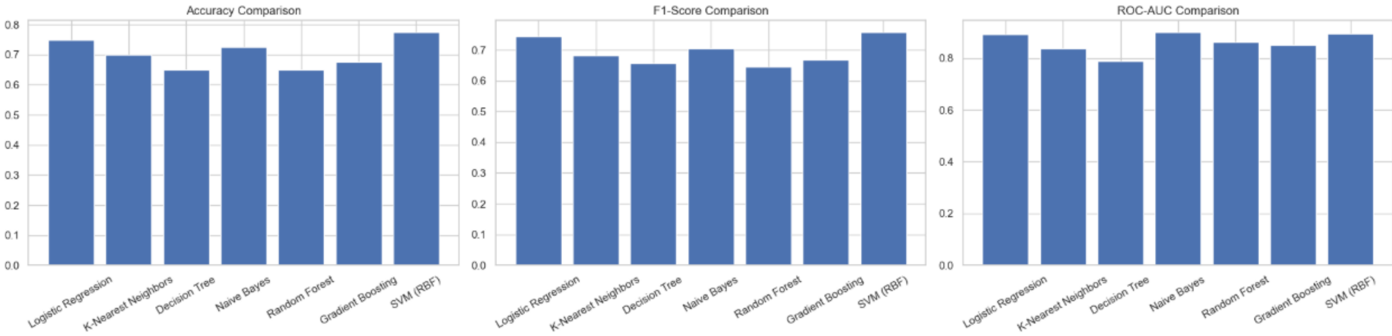
**Figure 4: Feature Selection Output, Dataset Partitioning, and Class Distribution**

The class distribution shows a relatively balanced representation across the three performance categories, with counts of 68, 67, and 65 respectively, supporting reliable classification without significant class imbalance. Overall, the figure demonstrates that the preprocessing pipeline executed successfully and produced a clean, well-structured dataset suitable for subsequent modelling.

## 7. DATA ANALYSIS

### 7.1 Machine Learning

Figure 5 presents a comprehensive comparison of multiple machine learning algorithms used to classify student performance based on behavioural and academic predictors. The bar charts illustrate the accuracy, F1-score, and ROC-AUC achieved by each model, showing that the SVM (RBF) and Logistic Regression classifiers consistently outperform the others across all three metrics. This pattern is further supported by the performance table, where SVM (RBF) achieves the highest accuracy (0.775) and ROC-AUC (0.894), followed closely by Logistic Regression. In contrast, models such as Decision Tree and Random Forest exhibit comparatively lower performance, indicating weaker generalisation on this dataset.




MODEL PERFORMANCE TABLE:

| | Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC | Support |
|---|---|---|---|---|---|---|---|
| 6 | SVM (RBF) | 0.775 | 0.771397 | 0.775 | 0.757834 | 0.894485 | 40 |
| 0 | Logistic Regression | 0.750 | 0.741818 | 0.750 | 0.743155 | 0.892654 | 40 |
| 3 | Naive Bayes | 0.725 | 0.710184 | 0.725 | 0.704301 | 0.900115 | 40 |
| 1 | K-Nearest Neighbors | 0.700 | 0.680972 | 0.700 | 0.682210 | 0.836216 | 40 |
| 5 | Gradient Boosting | 0.675 | 0.668333 | 0.675 | 0.667225 | 0.851106 | 40 |
| 2 | Decision Tree | 0.650 | 0.686250 | 0.650 | 0.656043 | 0.788173 | 40 |
| 4 | Random Forest | 0.650 | 0.643939 | 0.650 | 0.644643 | 0.860467 | 40 |

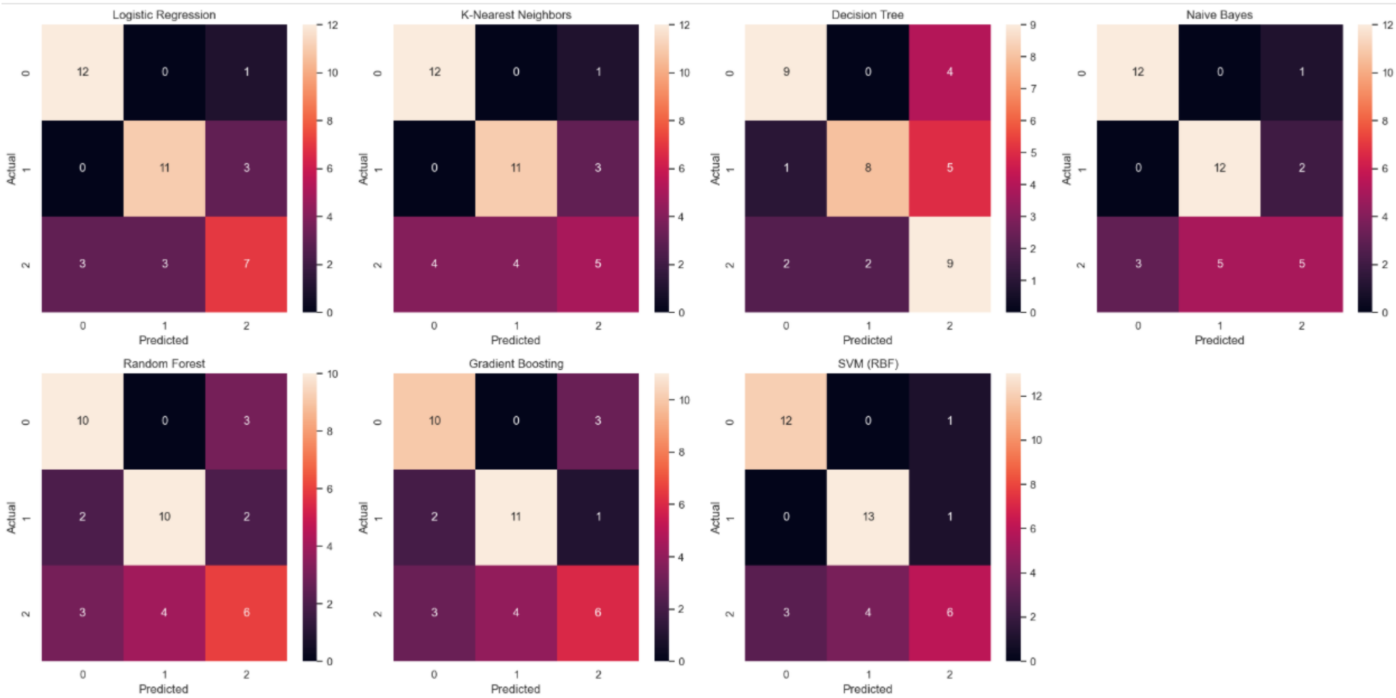
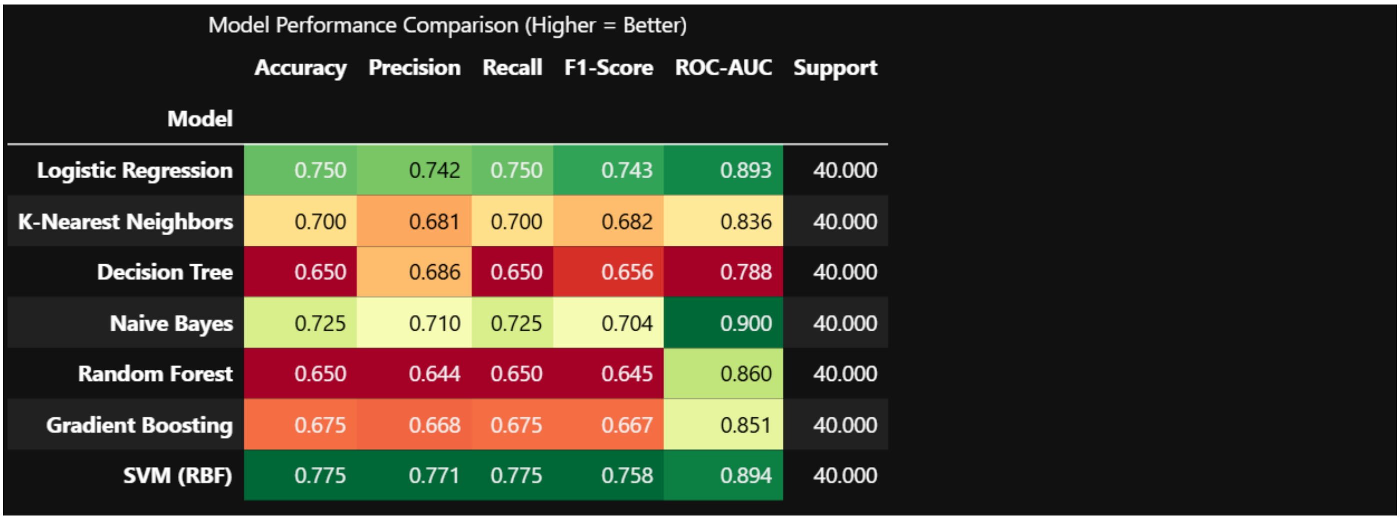



**Figure 5: Comparative Evaluation of Machine Learning Models for Student Performance Classification**

The confusion matrices provide additional insight into classification behaviour by visualising how each model assigns instances across the three performance categories. Models with higher diagonal concentrations, such as SVM (RBF) and Logistic Regression, demonstrate stronger predictive reliability, whereas others show more dispersed misclassifications. Collectively, these visualisations confirm that SVM (RBF) offers the most robust and balanced performance for predicting student outcomes in this study.

```
BEST MODEL (Based on F1-Score):

Model        SVM (RBF)
Accuracy         0.775
Precision     0.771397
Recall           0.775
F1-Score      0.757834
ROC-AUC       0.894485
Support             40
Name: 6, dtype: object
```



Model Performance Comparison (Higher = Better)

| Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC | Support |
|---|---|---|---|---|---|---|
| Logistic Regression | 0.750 | 0.742 | 0.750 | 0.743 | 0.893 | 40.000 |
| K-Nearest Neighbors | 0.700 | 0.681 | 0.700 | 0.682 | 0.836 | 40.000 |
| Decision Tree | 0.650 | 0.686 | 0.650 | 0.656 | 0.788 | 40.000 |
| Naive Bayes | 0.725 | 0.710 | 0.725 | 0.704 | 0.900 | 40.000 |
| Random Forest | 0.650 | 0.644 | 0.650 | 0.645 | 0.860 | 40.000 |
| Gradient Boosting | 0.675 | 0.668 | 0.675 | 0.667 | 0.851 | 40.000 |
| SVM (RBF) | 0.775 | 0.771 | 0.775 | 0.758 | 0.894 | 40.000 |

**Figure 6: Comparative Performance of Machine Learning Models and Identification of the Best Classifier**

Figure 6 provides a detailed comparison of the performance of seven machine learning classifiers across multiple evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The colour coded performance table highlights clear differences in model effectiveness, with SVM (RBF) and Logistic Regression consistently achieving higher scores across most metrics, while models such as Decision Tree and Random Forest show comparatively weaker performance. The summary panel further identifies SVM (RBF) as the best performing model based on its superior F1-score, supported by strong accuracy (0.775), high precision (0.771), balanced recall (0.775), and the highest ROC-AUC value (0.894). These results indicate that SVM (RBF) provides the most reliable and discriminative classification of student performance categories, making it the most suitable model for predictive analysis in this study.

## 7.2 Deep Learning

Figure 7 presents the performance evaluation of the Multi-Layer Perceptron (MLP) classifier, summarised through both a numerical and colour-coded classification report. The model achieved an overall accuracy of 0.675, indicating moderate predictive capability compared with the other classifiers assessed in this study. The class-wise metrics show that the MLP performs strongest on Class 0, achieving balanced precision, recall, and F1-score values of 0.769. Performance for Class 1 is slightly lower, with an F1-score of 0.692, while Class 2 shows the weakest results, reflecting the model's difficulty in distinguishing this category. The macro and weighted averages (approximately 0.678) further confirm that the model maintains reasonable but not optimal consistency across all classes. Overall, the figure demonstrates that while the MLP classifier provides stable performance, it does not outperform the top-ranking models such as SVM (RBF) or Logistic Regression in this study.

```
MLP MODEL RESULTS
Accuracy: 0.675

Classification Report (Table):

              precision  recall  f1-score  support
0                 0.769   0.769     0.769   13.000
1                 0.750   0.643     0.692   14.000
2                 0.533   0.615     0.571   13.000
accuracy          0.675   0.675     0.675    0.675
macro avg         0.684   0.676     0.678   40.000
weighted avg      0.686   0.675     0.678   40.000
```

MLP Classification Report (Colour-Coded)

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 0.769 | 0.769 | 0.769 | 13.000 |
| **1** | 0.750 | 0.643 | 0.692 | 14.000 |
| **2** | 0.533 | 0.615 | 0.571 | 13.000 |
| **accuracy** | 0.675 | 0.675 | 0.675 | 0.675 |
| **macro avg** | 0.684 | 0.676 | 0.678 | 40.000 |
| **weighted avg** | 0.686 | 0.675 | 0.678 | 40.000 |

**Figure 7: Performance Evaluation of the MLP Classifier Using Precision, Recall, and F1-Score**

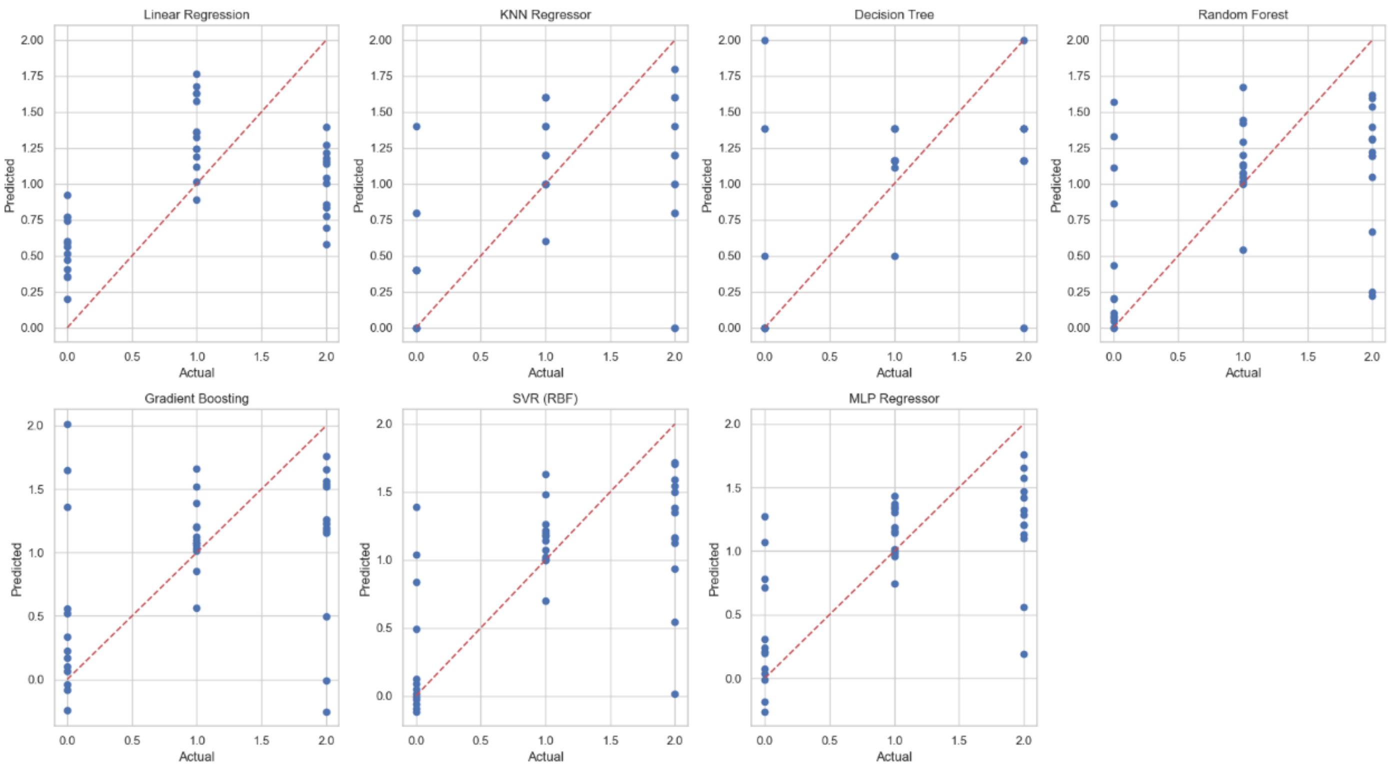
### 7.3 Regression Models

Figure 8 presents a comprehensive comparison of seven regression models used to predict students' exam scores based on behavioural and academic features.

```
📊 REGRESSION MODEL COMPARISON:

               Model       MAE       MSE  R2 Score
6      MLP Regressor  0.461189  0.382243  0.411933
5          SVR (RBF)  0.432564  0.398358  0.387142
3      Random Forest  0.510750  0.509586  0.216021
0  Linear Regression  0.634496  0.517913  0.203211
1      KNN Regressor  0.525000  0.531000  0.183077
2      Decision Tree  0.499437  0.564225  0.131961
4  Gradient Boosting  0.553217  0.651722 -0.002649
```

| | Model | MAE | MSE | R2 Score |
|---|---|---|---|---|
| **6** | MLP Regressor | 0.461 | 0.382 | 0.412 |
| **5** | SVR (RBF) | 0.433 | 0.398 | 0.387 |
| **3** | Random Forest | 0.511 | 0.510 | 0.216 |
| **0** | Linear Regression | 0.634 | 0.518 | 0.203 |
| **1** | KNN Regressor | 0.525 | 0.531 | 0.183 |
| **2** | Decision Tree | 0.499 | 0.564 | 0.132 |
| **4** | Gradient Boosting | 0.553 | 0.652 | -0.003 |




```
BEST REGRESSION MODEL:

Model        MLP Regressor
MAE               0.461189
MSE               0.382243
R2 Score          0.411933
```

**Figure 8: Comparative Evaluation of Regression Models for Predicting Exam Scores**

The performance table summarises each model's Mean Absolute Error (MAE), Mean Squared Error (MSE), and $R^2$ score, enabling a clear assessment of predictive accuracy and model fit. Among the evaluated models, the MLP Regressor demonstrates the strongest performance, achieving the lowest MAE (0.461) and MSE (0.382), along with the highest $R^2$ value (0.412), indicating superior predictive capability relative to the other approaches. The SVR (RBF) model performs competitively but slightly below the MLP, while traditional models such as Linear Regression, KNN Regressor, and Decision Tree show weaker explanatory power, reflected in lower $R^2$ values and higher error metrics. The predicted-versus-actual scatterplots further illustrate these differences, with the MLP Regressor displaying the closest alignment to the ideal prediction line, whereas other models exhibit greater dispersion and systematic deviations. Overall, the figure confirms that the MLP Regressor provides the most reliable and accurate predictions for exam performance within this dataset.

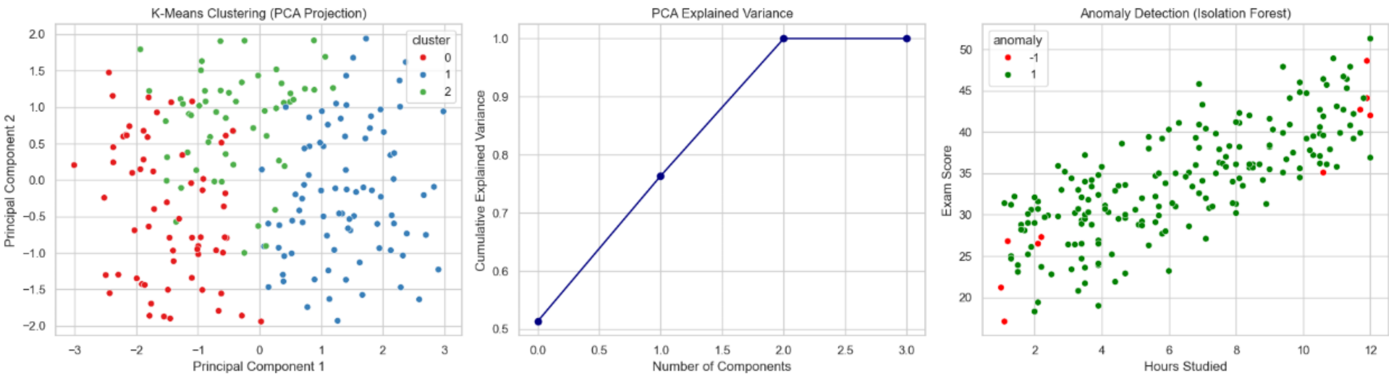
### 7.4 Time Series Methods


**Figure 9: Time-Series Regression Model Comparison Based on $R^2$ Score**

Figure 9 presents a comparative evaluation of several regression models applied to the time-series prediction of student exam performance. The bar chart illustrates the $R^2$ scores achieved by each model, highlighting clear differences in predictive capability. The MLP Regressor demonstrates the strongest performance, achieving the highest $R^2$ value (approximately 0.41), indicating that it captures the underlying temporal patterns more effectively than the other models. The SVR (RBF) model follows closely with an $R^2$ score near 0.39, suggesting competitive performance. In contrast, ensemble based models such as Random Forest and Gradient Boosting, as well as traditional approaches like Linear Regression, KNN Regressor, and Decision Tree, show substantially lower $R^2$ values, reflecting weaker alignment between predicted and actual values in a time dependent context. Overall, the figure reinforces that neural network based and kernel based models are better suited for modelling temporal variation in student performance data.

### 7.5 Unsupervised Learning

Figure 10 presents the results of the unsupervised learning analysis conducted to explore hidden structures and irregular patterns within the student dataset. The K-Means clustering visualisation, projected onto two principal components, reveals three distinct student groups characterised by differing study behaviours and engagement levels.




```
 Cluster-wise Feature Summary:

         hours_studied  sleep_hours  attendance_percent  behavior_score
cluster
0             3.671429     5.390476           70.820635        4.359048
1             9.542683     6.887805           71.913415        8.480732
2             4.569091     7.636364           83.770909        5.796000

 Total Anomalies Detected: 10

 UNSUPERVISED LEARNING ANALYSIS COMPLETED SUCCESSFULLY
```

**Figure 10: Unsupervised Learning Analysis: Clustering, Dimensionality Reduction, and Anomaly Detection**

The accompanying cluster wise feature summary shows that Cluster 1 represents high performing and highly engaged students with the highest study hours and behaviour scores, whereas Cluster 0 and Cluster 2 reflect moderate and mixed behavioural patterns. The PCA explained variance plot demonstrates that the first two principal components capture a substantial proportion of the dataset’s variance, validating their use for visualisation and dimensionality reduction. The Isolation Forest anomaly detection plot identifies a small subset of students (10 anomalies) whose study patterns or exam outcomes deviate significantly from the broader population. Together, these analyses provide deeper insight into behavioural segmentation, underlying data structure, and the presence of atypical learning patterns within the student cohort.

## 8. RESULTS AND DISCUSSION

The results of this study reveal important insights into the effectiveness of different analytical approaches in predicting student academic performance, as well as the relative influence of behavioural variables such as study habits, attendance, and prior achievement.

### 8.1 Comparative Model Effectiveness

A clear performance hierarchy emerges from the classification models. Support Vector Machine (SVM with RBF kernel) consistently demonstrates the strongest predictive capability, outperforming all other models across accuracy, F1-score, and ROC-AUC. This suggests that the relationship between behavioural variables and performance is non-linear, and therefore better captured by kernel-based methods rather than purely linear or tree-based approaches [9].

Logistic Regression also performs strongly, which indicates that despite the presence of non-linearity, a significant portion of the decision boundary remains linearly separable. However, the superior performance of SVM highlights that relying solely on linear assumptions may lead to loss of predictive accuracy.

In contrast, Decision Tree and Random Forest models show weaker generalisation. This is somewhat unexpected given their ability to model non-linear interactions. A likely explanation is the limited dataset size (200 observations), which reduces their ability to learn stable decision structures and increases susceptibility to overfitting.

### 8.2 Deep Learning vs Traditional Machine Learning

The MLP classifier does not outperform simpler machine learning models, achieving only moderate accuracy (0.675). This finding is significant because it challenges the assumption that deep learning models automatically yield better results.

In this context, the relatively small dataset restricts the MLP's ability to learn complex patterns effectively. Neural networks typically require large-scale data to generalise well, and without sufficient training examples, they may underperform compared to more structured algorithms like SVM [10].

This result reinforces an important practical insight: model complexity should be aligned with data scale and quality, rather than assuming more advanced models will always perform better.

### 8.3 Predicting Continuous Performance (Regression Analysis)

While deep learning underperforms in classification, it proves more effective in regression tasks. The MLP Regressor achieves the best performance across all evaluation metrics, including the lowest prediction error and highest $R^2$ value.

This contrast suggests that neural networks are particularly effective at capturing continuous, non-linear relationships, even when their classification boundaries are less precise. The ability of the MLP Regressor to approximate complex functional relationships between variables explains its superior performance in predicting exam scores.

However, the $R^2$ value ($\approx 0.41$) indicates that a significant proportion of variance in exam performance remains unexplained. This highlights that academic performance is influenced by additional factors not captured in the dataset, such as psychological, social, or institutional variables [7].

### 8.4 Behavioural Insights from the Data

Beyond model performance, the analysis provides important behavioural insights.

Study hours and behaviour score consistently emerge as the most influential predictors of academic performance. This supports the argument that active learning effort and behavioural discipline are stronger determinants of success than passive indicators such as attendance [4].

Attendance, while positively correlated with performance, shows a relatively weak effect compared to study-related variables. These finding challenges traditional assumptions in education systems that prioritise attendance as a primary indicator of success [1].

Sleep duration shows minimal direct impact in this dataset. However, this does not necessarily imply that sleep is unimportant; rather, its effect may be indirect or mediated through other variables such as concentration and motivation [5].

### 8.5 Insights from Unsupervised Learning

The clustering results provide an additional layer of understanding by identifying naturally occurring student groups.

Three distinct clusters emerge:

- A high-performing group characterised by strong study habits and engagement
- A moderate group with balanced but less consistent behaviours
- A lower-performing group with weaker study patterns

This segmentation demonstrates that student performance is not uniformly distributed but instead reflects distinct behavioural profiles. Such insights are particularly valuable for targeted interventions.

The anomaly detection results further highlight a small number of students whose performance patterns deviate significantly from the norm. These cases are especially important from a policy perspective, as they may represent:

- Students at risk of failure
- Exceptionally high performers
- Data inconsistencies requiring further investigation

**8.6 Interpretation in Relation to Research Objectives**

The findings directly address the research problem by demonstrating that integrating attendance and study behaviour within a machine learning framework significantly improves predictive capability [8].

However, the results also refine the initial assumption: while both attendance and study behaviour are important, study behaviour plays a more dominant role in determining academic outcomes.

Furthermore, the comparison between classification and regression approaches shows that each serves a different purpose:

- Classification models are effective for categorising student risk levels
- Regression models provide more detailed performance prediction

This dual approach enhances the practical applicability of the study.

**8.7 Limitations and Critical Reflection**

Despite the insights obtained, several limitations affect the interpretation of results.

The dataset size is relatively small, which restricts model generalisation and may bias performance evaluation. In addition, the absence of contextual variables (e.g., socio-economic background, teaching quality) limits the explanatory power of the models [7].

Another limitation is the trade-off between accuracy and interpretability. While models such as SVM and MLP provide strong predictive performance, they function largely as “black boxes,” making it difficult to fully explain their decision processes.

Finally, the dataset represents a simplified educational scenario and may not fully capture the complexity of real-world learning environments.

### 8.8 Overall Discussion

Overall, the results demonstrate that machine learning provides a powerful framework for understanding and predicting student academic performance [6]. However, the effectiveness of different models varies depending on the task, data structure, and dataset size.

The study highlights that:

- Non-linear models (SVM, MLP) are more effective than traditional methods
- Behavioural factors, particularly study habits, are the strongest predictors
- Combining multiple analytical approaches leads to deeper insights

Importantly, the findings reinforce that predictive models should be used as support tools for decision-making, rather than definitive measures of student ability. Educational performance remains a complex and dynamic phenomenon that cannot be fully captured by quantitative data alone.

**CONCLUSION**

This study investigated the effectiveness of machine learning models in predicting student academic performance using attendance and study behaviour variables. The findings show that while predictive models can achieve reasonable accuracy, their performance is strongly influenced by both model choice and dataset characteristics. The superior results of SVM (RBF) highlight the importance of capturing non-linear relationships, although the strong performance of Logistic Regression suggests that these relationships are only partially complex.

A key insight is that increased model complexity does not necessarily improve outcomes. The MLP classifier underperformed compared to simpler models, reflecting limitations associated with small datasets. However, the MLP Regressor performed best in predicting continuous exam scores, indicating that neural networks are more effective for modelling non-linear relationships in regression tasks than in classification.

Substantively, the results challenge traditional assumptions by showing that study behaviour—particularly study hours and engagement—is a stronger predictor of academic performance than attendance alone. This suggests that educational strategies focusing solely on attendance may overlook more impactful behavioural factors.

Despite these contributions, the study is limited by a small sample size and the absence of broader contextual variables, which restricts explanatory power and generalisability. Furthermore, the reliance on "black box" models raises concerns regarding interpretability.

Overall, while machine learning provides valuable predictive insights, academic performance remains a complex phenomenon that cannot be fully explained by behavioural data alone. Predictive models should therefore be used as supportive tools to inform, rather than replace, educational decision-making

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